PRISM-ANT: The Core Algorithm

Simple explanation of how ANT-EV splits charging power

PRISM-ANT is the heart of ANT-EV. It decides how the station splits power. It does not simply choose the highest-priority vehicle. It first gives every vehicle the minimum safe power needed to meet its deadline, then shares all remaining safe power fairly.

What PRISM-ANT means

Predictive **R**eadiness, **I**ntegrity, **S**afety **M**asking **- A**daptive **N**ecessity **T**iering.

The simple 5-step decision

Step	What it does	Why it matters
1. Read	Read SOC, target, deadline, temperature, vehicle limit, station limit, emergency flag, and model outputs.	The station has the current facts.
2. Safety mask	Find the maximum safe power for each EV. Temperature, predicted temperature, vehicle and station limits are hard limits.	No priority can break a safety limit.
3. Readiness reserve	Calculate the minimum kW required to deliver the needed energy before the deadline.	A vehicle close to departure is not treated like a vehicle with plenty of time.
4. Fair sharing	After reserves, divide every remaining safe kW by priority.	The station stays fully utilised when safe demand exists.
5. Explain and repeat	Record the cap, reserve, model signals and assigned power. Repeat on arrival, completion, or state change.	The decision is explainable and dynamic.

The two power tiers

Tier 1 - Safe readiness reserve: reserve_kW = energy_needed_kWh / hours_until_departure / charging_efficiency, limited by the hard safety cap. If reserve demand is greater than station capacity, PRISM-ANT rations the reserve by transparent priority.

Tier 2 - Fair remaining-power share: distribute the remaining capacity by the PRISM priority score, never exceeding an EV cap.

Model inputs used by PRISM-ANT

Model	Parameter sent to PRISM-ANT	Use
M1	energy_need_kwh, readiness risk, confidence	Calculates Tier 1 reserve and deadline pressure.
M2	forecast pressure	Raises priority only when the EV also has deadline risk.
M3	health risk, confidence	Protects normal battery life and can lower a normal-session cap.
M4	thermal risk, safe power cap	Creates a hard temperature/C-rate cap.
M5	future temperature risk	Prevents a power request that becomes unsafe later.
M6	strategy value	Ranks safe choices; it cannot re-enable an unsafe choice.

Priority score - plain explanation

Priority combines: urgency + deadline risk + energy need + strategy value + low flexibility + forecast pressure when paired with deadline risk - normal-session battery-health protection. Emergency adds urgency, but it cannot override the safety mask.

Why it is stronger than common approaches

Approach	Limitation	PRISM-ANT improvement
First come, first served	Early arrivals can consume power even when another EV will miss a deadline.	Uses a calculated readiness reserve.
Equal split	Every EV gets the same kW even with different needs and safe caps.	Uses safe caps, need, deadline, and fair remaining-power sharing.
One priority score	Safety may be mixed into the score and become negotiable.	Safety is applied first as a non-bypassable mask.
Generic dynamic allocation	May show a final kW without showing why.	Stores model signal, cap, reserve, priority, and command for every EV.

Important boundary

This is a software allocation method. Real charging still requires vehicle BMS limits, charger controls, electrical protection, telemetry validation, and safety review. It is not a guarantee against fire or battery damage.

PRISM-ANT Feature Checklist

This page lists the complete implemented algorithm features in one place.

Feature	What it does
Safety-first allocation	Creates a hard cap from current temperature, predicted temperature, vehicle, charger, and station limits before any priority is calculated.
Deadline readiness reserve	Calculates the minimum safe kW needed to meet the requested target before departure.
Two-tier allocation	Allocates reserves first, then shares remaining safe capacity fairly.
Full safe utilisation	Continues allocation until the station capacity or every safe EV cap is reached.
M1-M6 signal contract	Uses energy need, forecast, health, thermal, future-state, and strategy outputs with named handoffs.
Forecast-deadline interlock	Uses station forecast only when the individual EV also has deadline risk.
Emergency rule	Raises urgency while preserving every physical safety limit.
Confidence-aware fallback	Uses conservative limits when predictive confidence is weak.
Explainable event record	Shows kW, cap, reserve, model signals, priority reason, and recalculates after live events.

Patent note

These features define the technical method. Patentability depends on whether the exact combination is new and non-obvious after a professional prior-art search.